

Qualified FPGA implementation evidence

Wide core top — negative result

Mapping: 763 logic elements
505 registers
151 external pins
Fitter: FAILED at 151 / 151 pins
No successful-fit timing claim

replace wide diagnostics with CSR

Compact CSR wrapper — fitted result

824 / 24,624 logic elements (3%)
505 registers
119 / 151 pins (79%)
Fitter: SUCCESS; zero virtual pins
Target: Cyclone 10 LP 10CL025YU256C8G

Why the wrapper changes feasibility

Wide debug/status buses consume package pins
CSR serializes diagnostics into byte reads
Logic cost rises slightly; pin demand falls by 32

Internal-clock timing

80 MHz (12.5 ns constraint)
worst setup slack: +1.013 ns
worst hold slack: +0.179 ns
reported negative slack: 0

Qualification

No board pin locations or external I/O delays → internal timing evidence only; not board closure, ASIC PPA, or UCle PHY performance